Towfish Proton Precession Magnetometer (PPM) Design Documentation

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ECE 455: CAPSTONE DESIGN IN ELECTRICAL AND COMPUTER ENGINEERING

Agenda

- Background and PPM Coil Design
- Circuit Block Diagram
- Circuit Schematics, PCB and BOM
- Software
- Physical Towfish
- Future Improvements

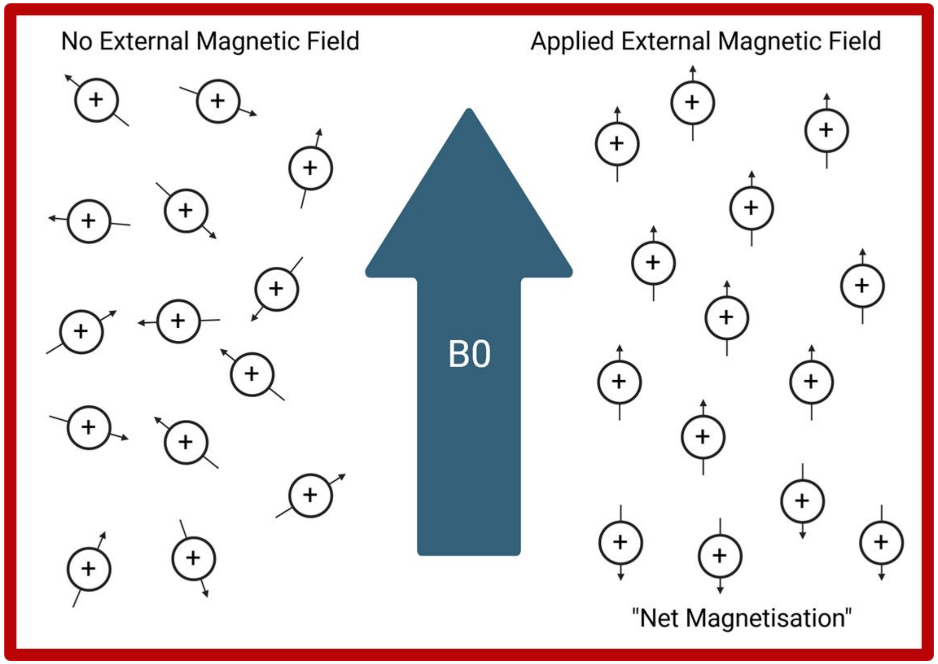
Background and Coil Design

Proton Precession Magnetometry Background

- **Polarization:** A DC current through copper coils generates a magnetic field (B_0) that aligns protons in a fluid such as water or kerosene.

- **Switch-off:** The current is abruptly cut off, causing the protons to begin realigning with Earth's weaker magnetic field.

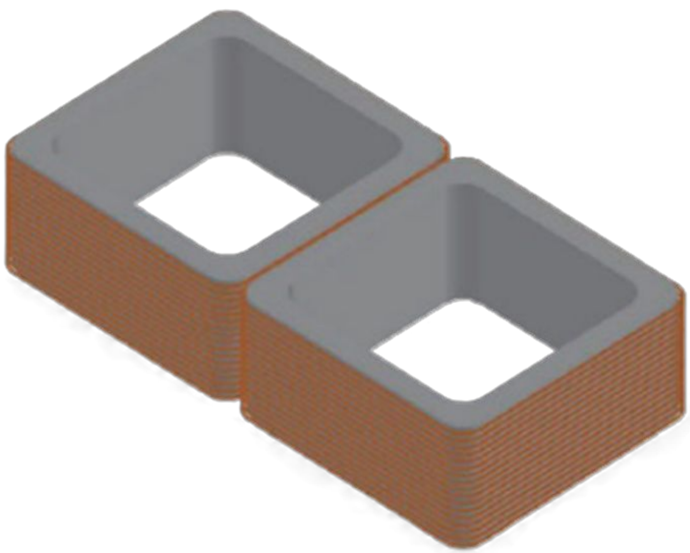
- **Induction & Measurement:** As the protons precess, they induce a small AC voltage in the coils, which is amplified and measured to determine the magnetic field strength.



Coil Design

GitHub 3D Model Link: Link

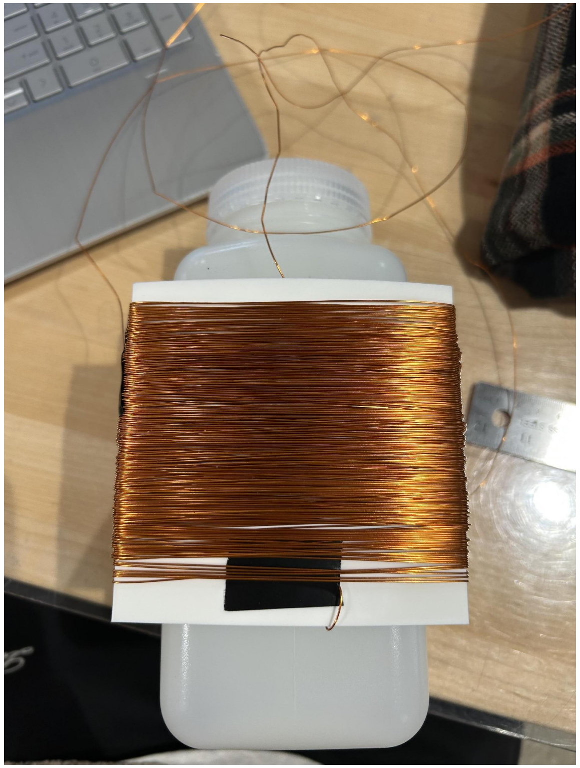
- 3D-printed hollow coil frame
- 10 x 10 x 10 cm
- 265 turns per coil, 530 total turns
- Oppositely Wound
- 18.5 Ohms Resistance



Coil Design Build Steps

- **Enclosure:** Custom 3D-printed 10x10x10 cm square housings
- **Configuration:** Oppositely wound coils, each wound to exactly 265 turns.
- **Material:** 24 AWG copper wire
- **Insulation:** Windings encapsulated in electrical tape to protect against shorts and improve mechanical stability
- **Mounting:** Coils securely mounted in PCB holder within Towfish
- **PCB Connectivity:** Direct terminal soldering to Coil(+) and Coil(-) pads.

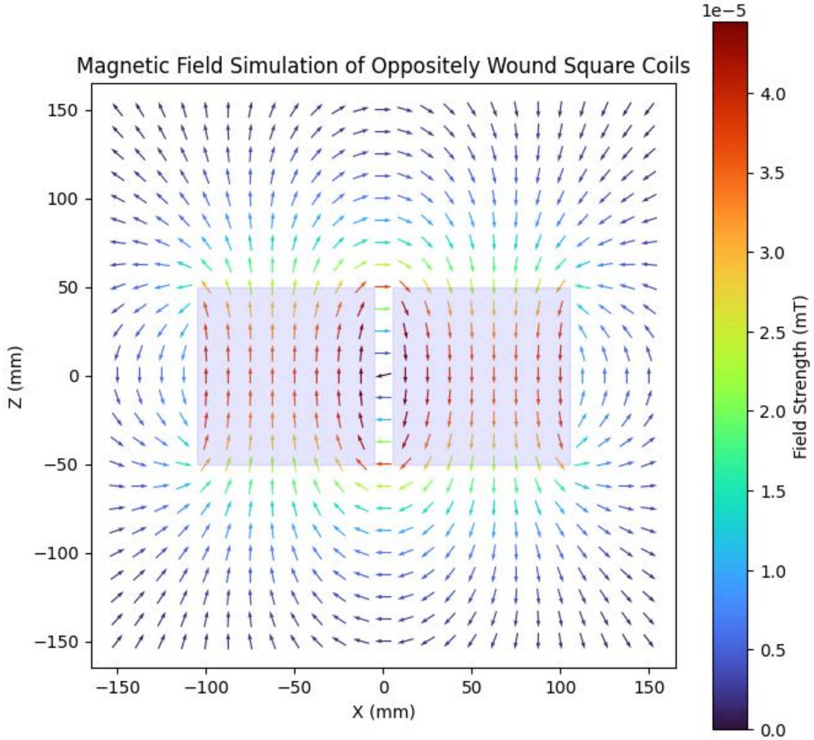
Image of Final Coils Design



Reasoning For Oppositely Wound Coil Design

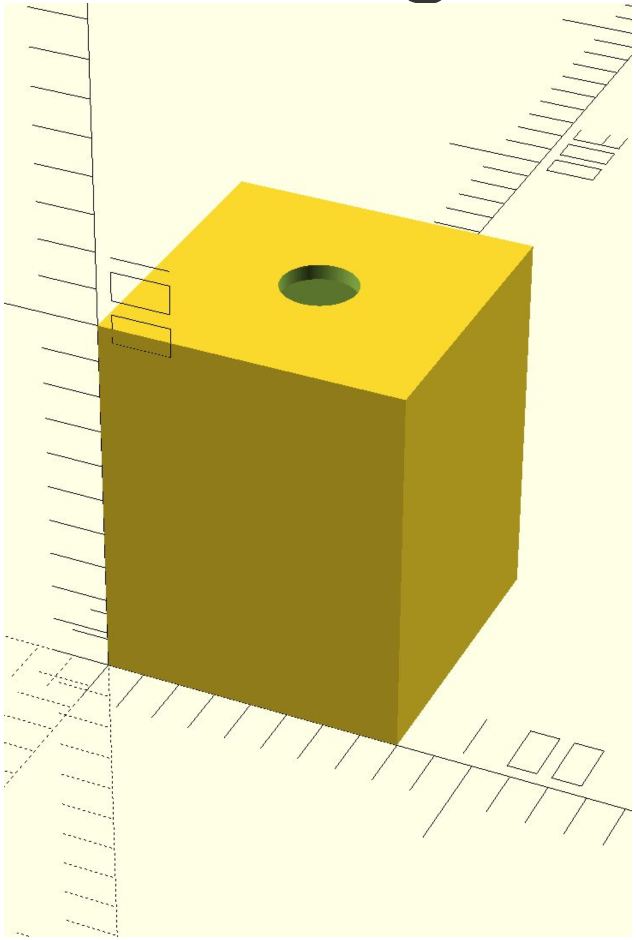
- Oppositely wound coils help eliminate external noise by cancelling voltage induced in one coil equally and oppositely with voltage induced in the other coil

- Flux Concentration: Oppositely wound coils compress the magnetic field into the center of the two coils, maximizing the polarization of the fluid



Proton Fluid Storage Container Design

- 3D Model of Proton Fluid Storage Container
  - Link
- Coated 3D prints with JB Marine Weld to ensure a leak proof seal and prevent kerosene leakage
  - J-B WELD Epoxy Adhesive: MarineWeld

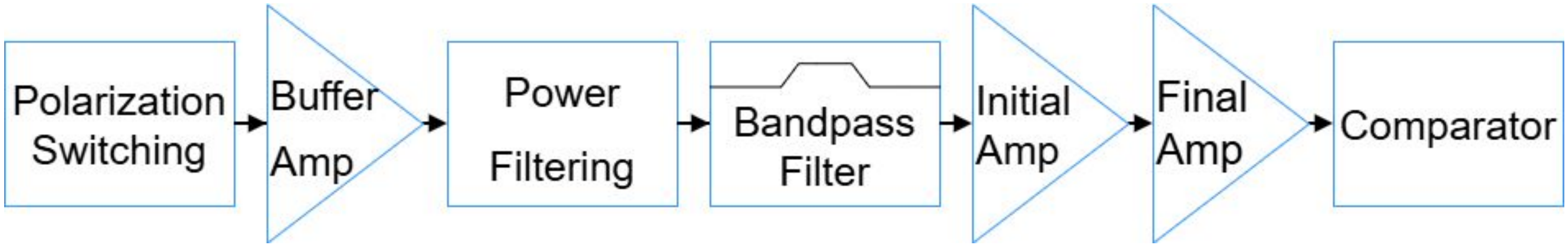


Proton Magnetometer Circuit Block Diagram

Proton Magnetometer Circuit Design Block Diagram

Design for signal path to accurately measure Proton precession signal

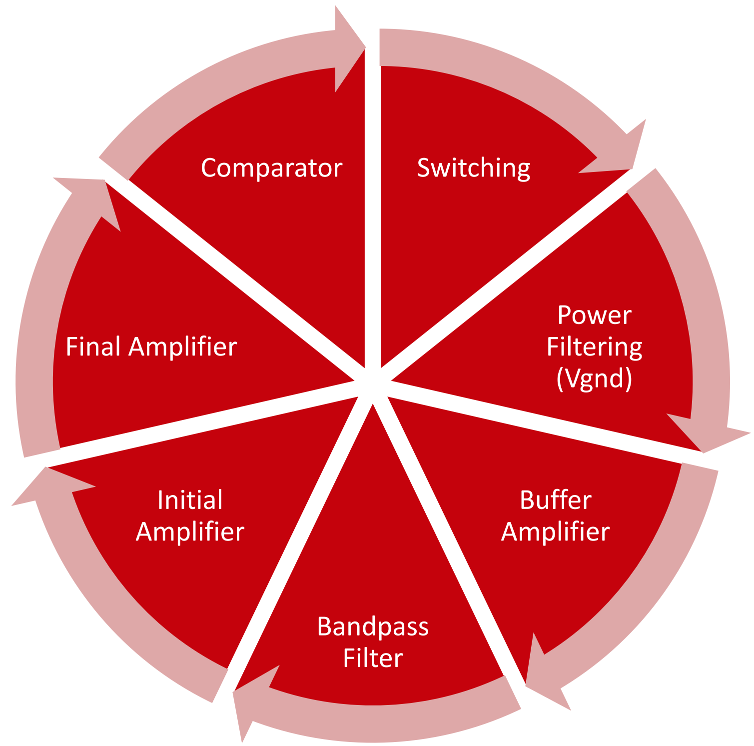
- Powers and controls proton alignment
- Amplifies the proton alignment signal induced after polarization



Proton Magnetometer Circuit Schematics

Proton Magnetometer Circuit Schematics

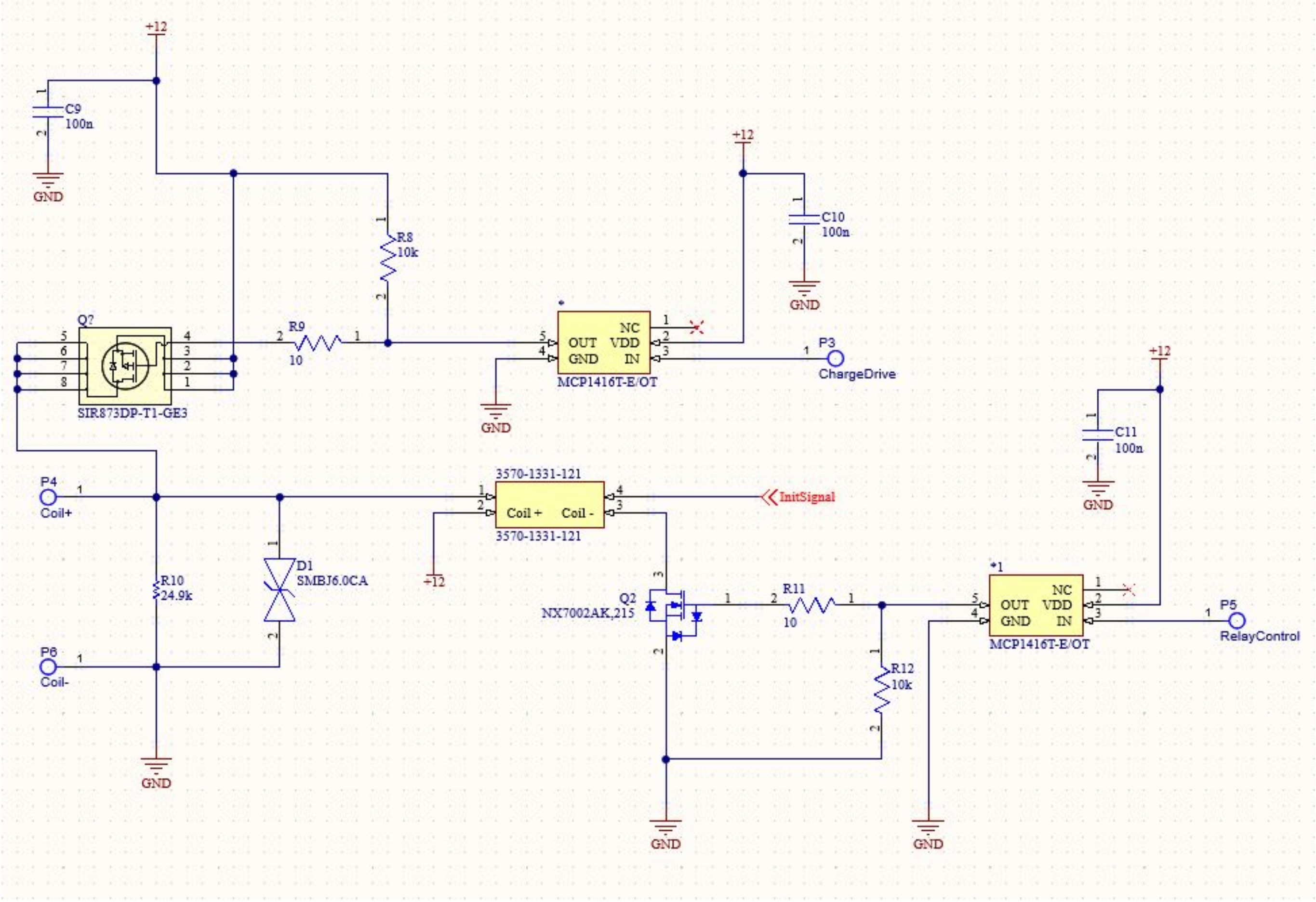
1. Switching Circuit
2. Power Filtering (Vgnd)
3. “Buffer” Amplifier
4. Bandpass Filter
5. Initial Amplifier
6. Final Amplifier
7. Comparator



Switching Circuit Schematic

- Link to Switching Circuit design files
  - [Link]
  - Simulation File
- Switching Circuit Description
  - Switches coil between DC polarization and AC sensing circuit to read signal
  - When charging is complete and MOSFET is turned off, TVS diode and resistor are used to deplete stored energy before sensing
  - Gate driver is used to control high-side MOSFET, and relay control MOSFET
    - Current PCB design and BOM includes non-inverting drivers, but currently an inverting driver is soldered to the board for the high side MOSFET, along with a pulldown resistor. Both work, just changes "on" signal from microcontroller.
    - Coil charge circuit is currently turned on with an input of 3.3V. Once saturated, turn off charge circuit, wait set time (~900us), then turn on sensing circuit relay with 3.3V input.
  - This circuit was chosen for simplicity, and rapid discharge of the coil
    - We recommend using a low side N-channel Mosfet to float the coil once it's discharged so you can use an instrumentation amplifier to remove noise in the coil before the main gain stages.

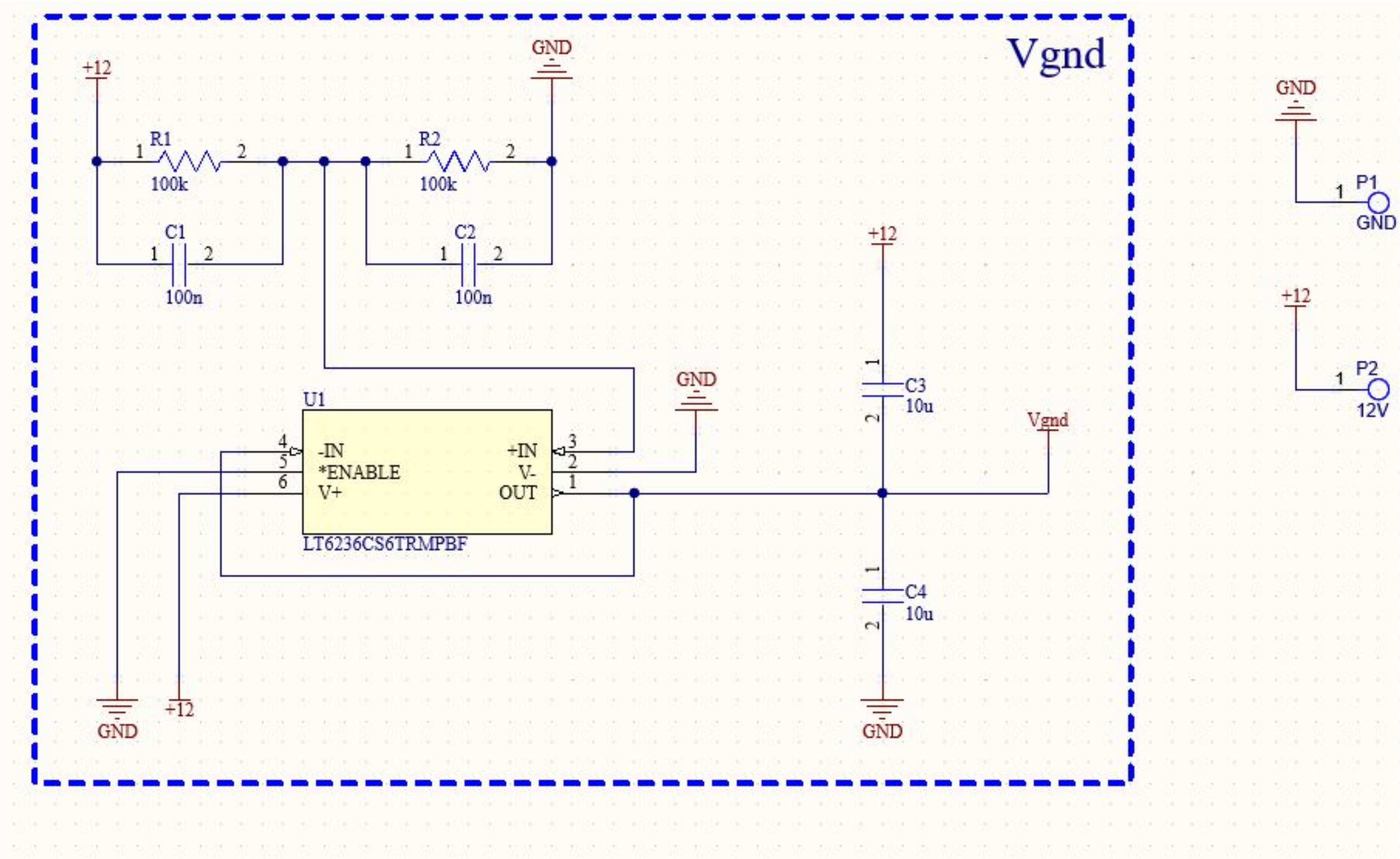
Switching Circuit Schematic Design



Power Filtering Circuit Schematic

- Link to Power Filtering Circuit design files
  - Link
- Power Filtering Circuit Description
  - Generates stable virtual ground for amplifiers
  - Voltage divider creates ~6V for a virtual ground with two resistors and decoupling capacitors
  - Buffer amplifier used to stabilize and strengthen virtual ground for use in circuit
- We chose to create a virtual ground instead of creating a negative voltage for op amps to reduce switching noise

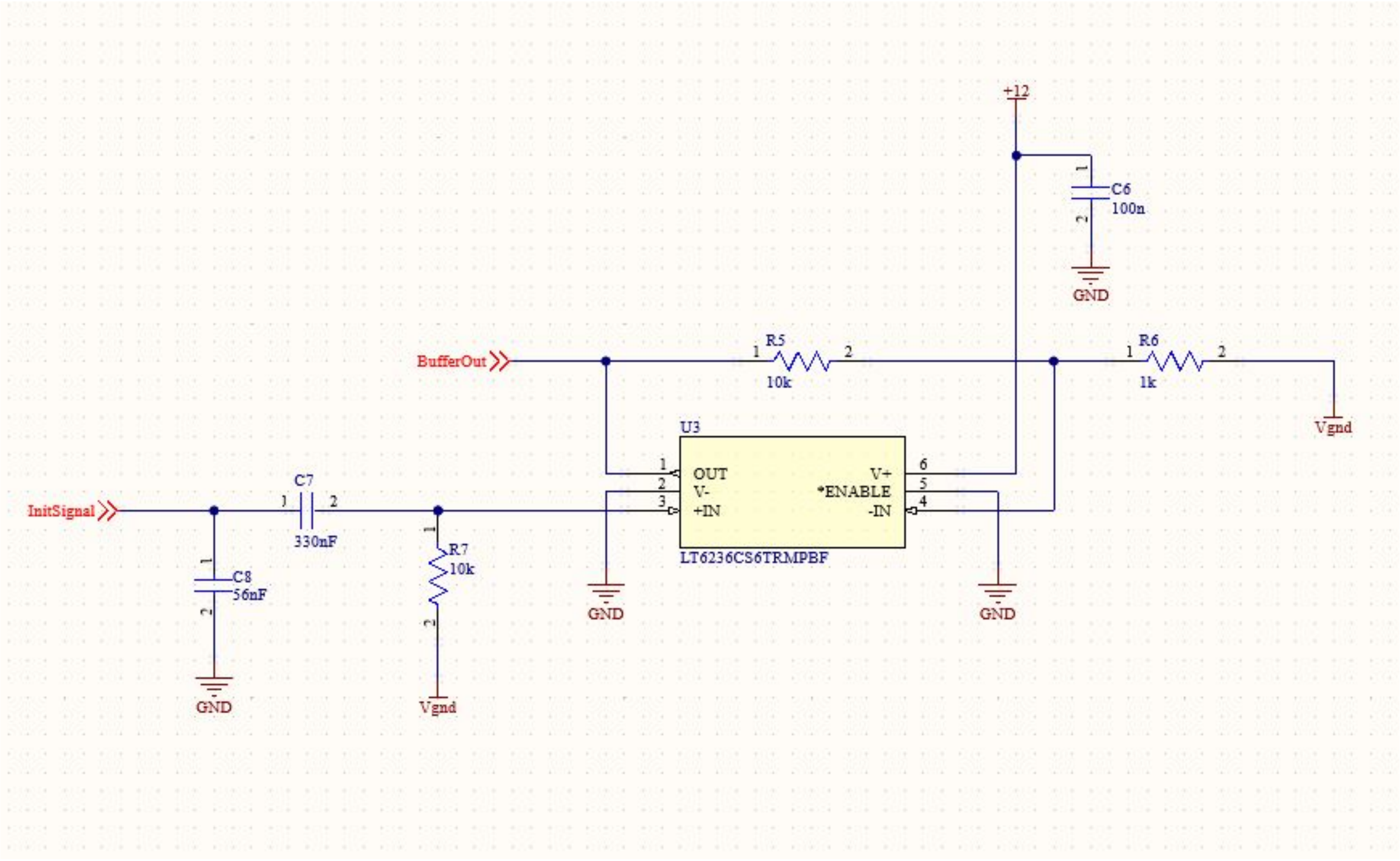
Power Filtering Circuit Schematic Design



“Buffer” Amplifier Circuit Schematic

- Link to Buffer Amplifier Circuit design files
  - Link
  - Simulation Link
- Buffer Amplifier Circuit Description
  - Chosen for high impedance input for signal to prevent source loading
  - Technically not a buffer stage as it has 10x gain
  - Before input, has a decoupling capacitor and a resistor to Vgnd to add DC offset to signal

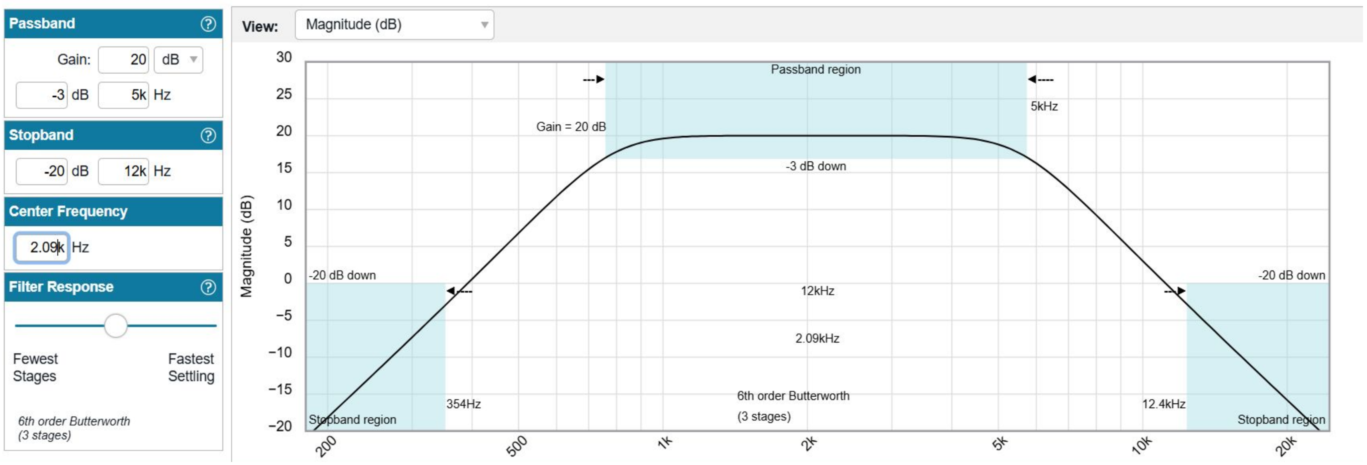
Buffer Amplifier Circuit Schematic Design



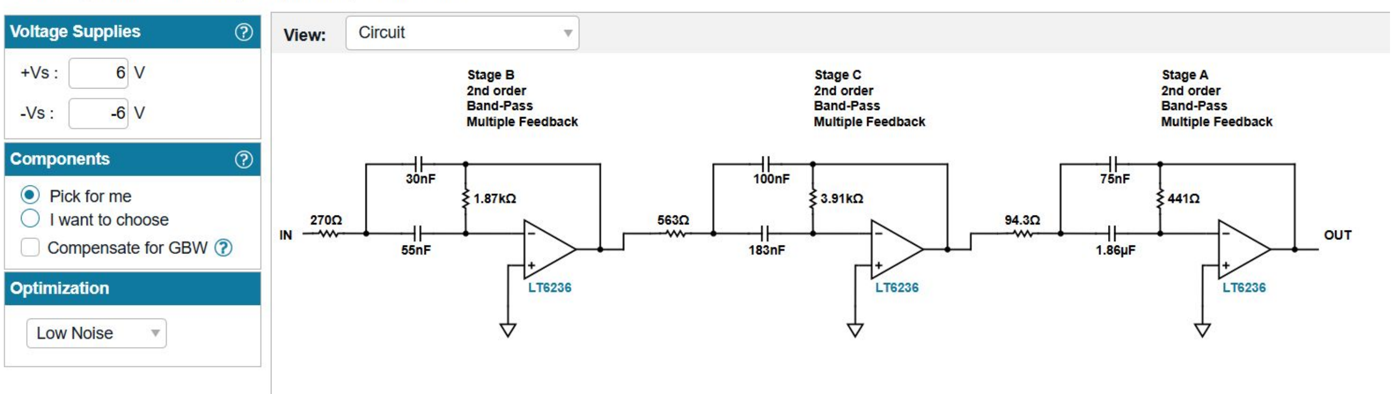
Bandpass Filter Circuit Schematic

- Link to Bandpass Filter Circuit design files
  - Link
- Bandpass Filter Circuit Description
  - Isolates target band (850–5kHz), adds 10x gain

Bandpass Circuit Design

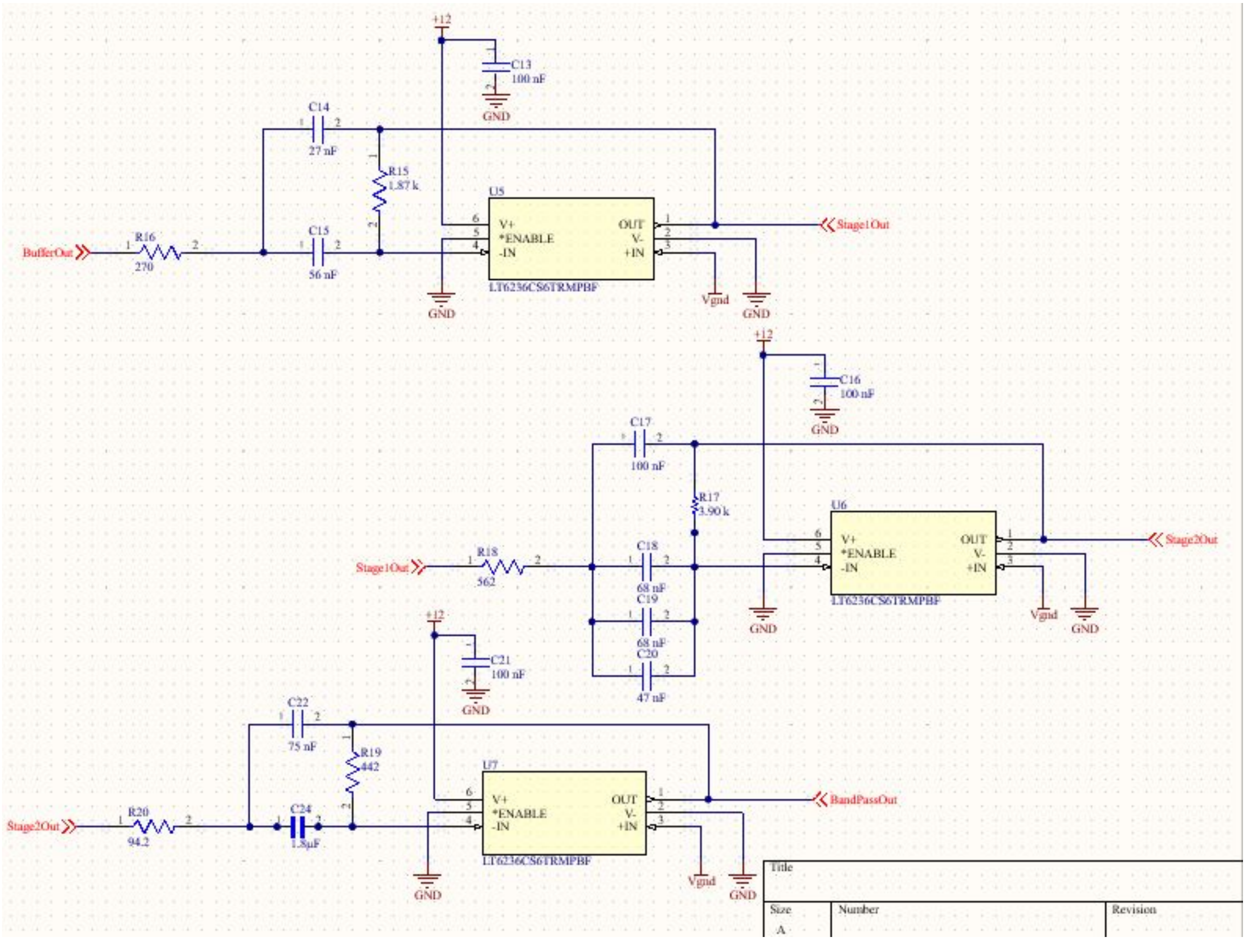


Band-Pass; 6th order Butterworth; 5kHz passband; 2.09kHz center



- Target band (850-5kHz)
- Utilized Low Noise Components
- Gain of 20 dB or 10x
- +/- 6V to be compatible with virtual ground

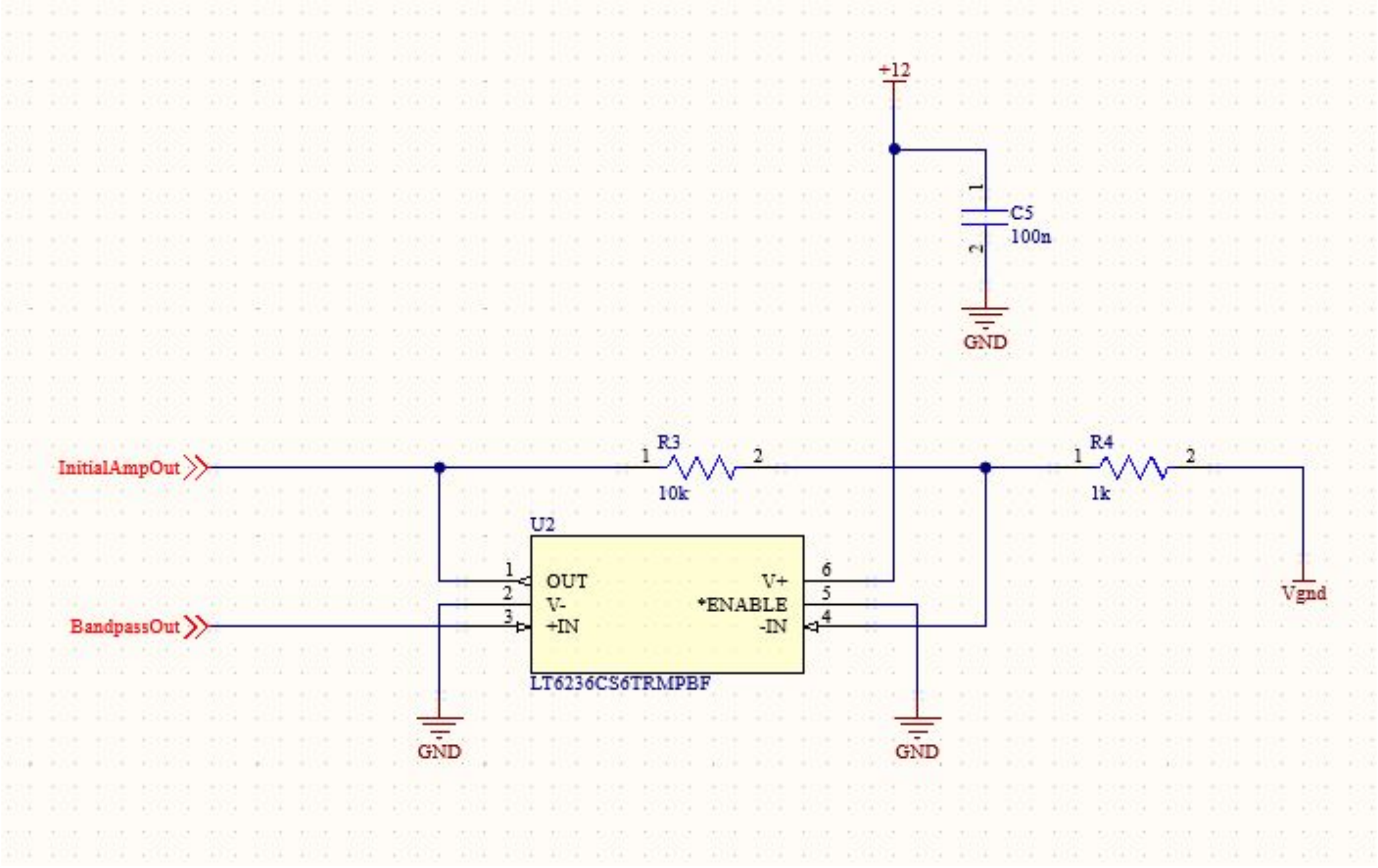
Bandpass Filter Circuit Schematic Design



Initial Amplifier Circuit Schematic

- Link to Initial Amplifier Circuit design files
  - [Link]
- Initial Amplifier Circuit Description
  - Boosts signal by 10x using for signal enhancement
  - Simple non-inverting amplifier
  - Current PCB set to a gain of 2x for noise testing
- Chosen for simplicity and reliability

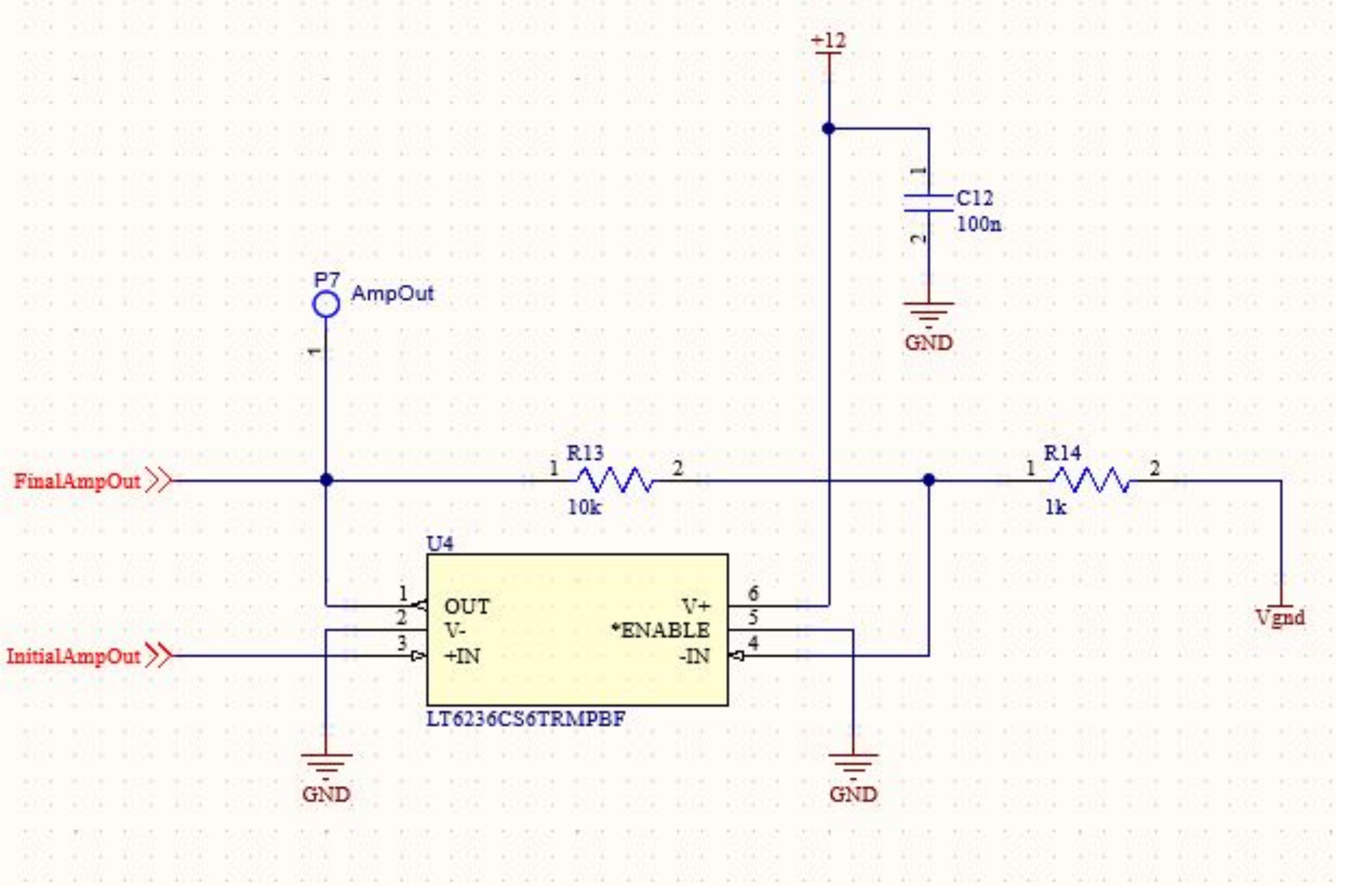
Initial Amplifier Circuit Schematic Design



Final Amplifier Circuit Schematic

- Link to Final Amplifier Circuit design files
  - Link
- Final Amplifier Circuit Description
  - Boosts signal by 10x for signal enhancement
  - Simple non-inverting amplifier, same as initial amplifier
  - Current PCB set to a gain of 2x for noise testing
- Chosen for simplicity and reliability

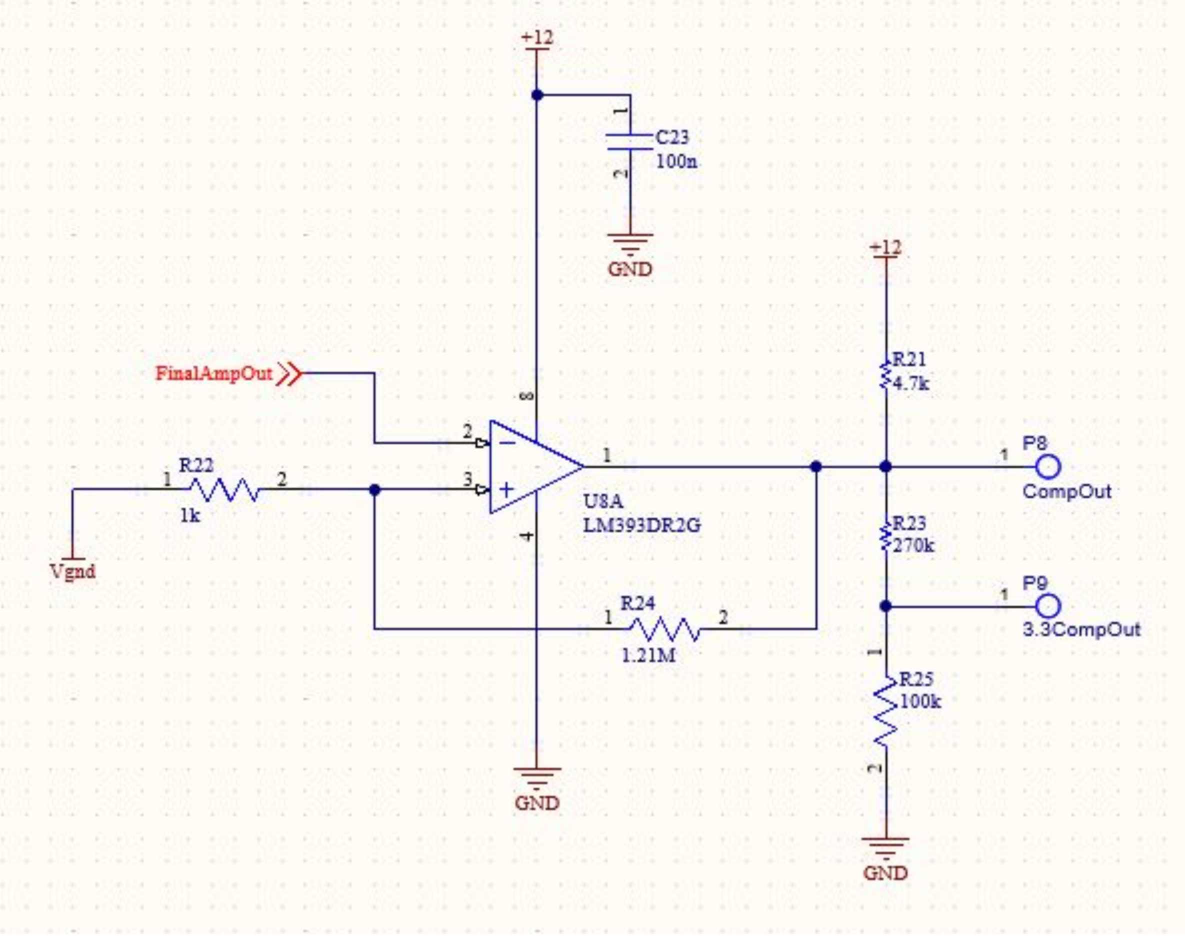
Final Amplifier Circuit Schematic Design



Comparator Circuit Schematic

- Link to Comparator Circuit design files
  - Link
  - Simulation File
- Comparator Circuit Description
  - Digitizes sine wave to a square wave for microcontroller reading
  - Hysteresis added with R24 and R22 to prevent chatter at zero crossings, adjustable if needed
  - Voltage divider to lower output to ~3.3V
- Chosen to simplify software frequency measurement

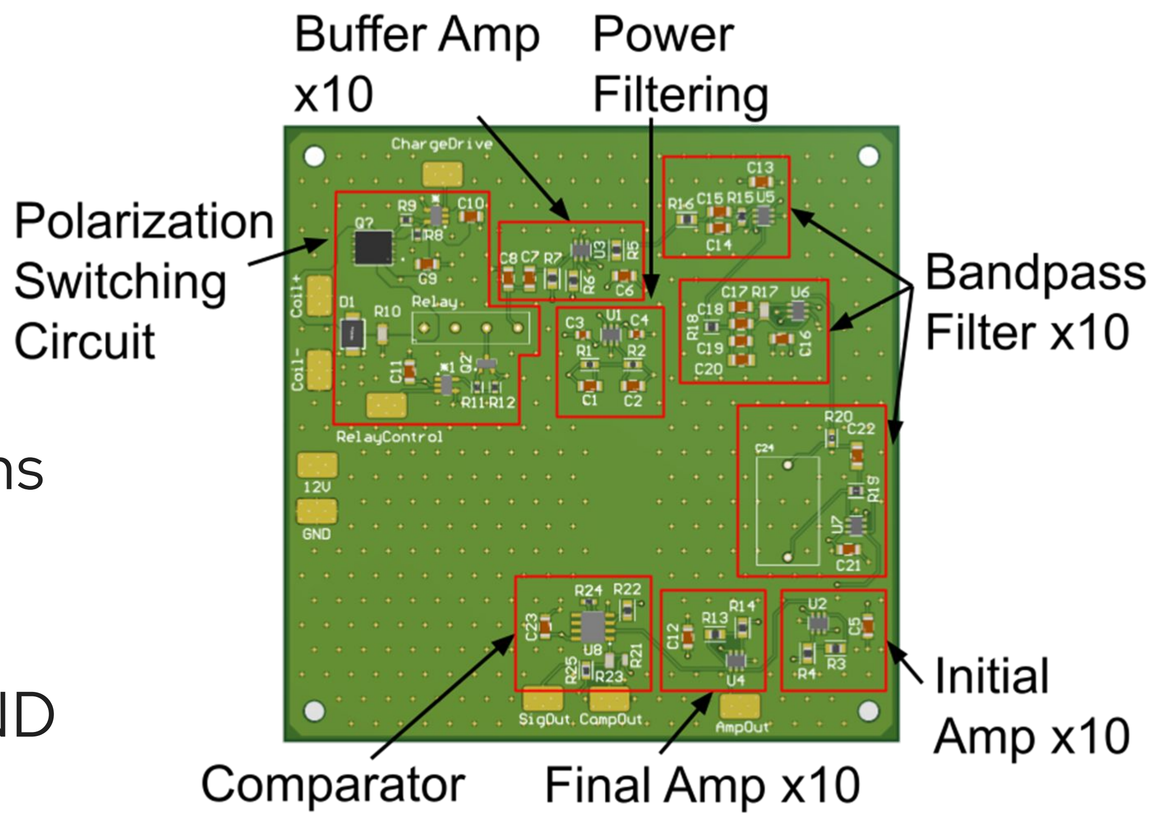
Comparator Circuit Schematic Design



PCB Design

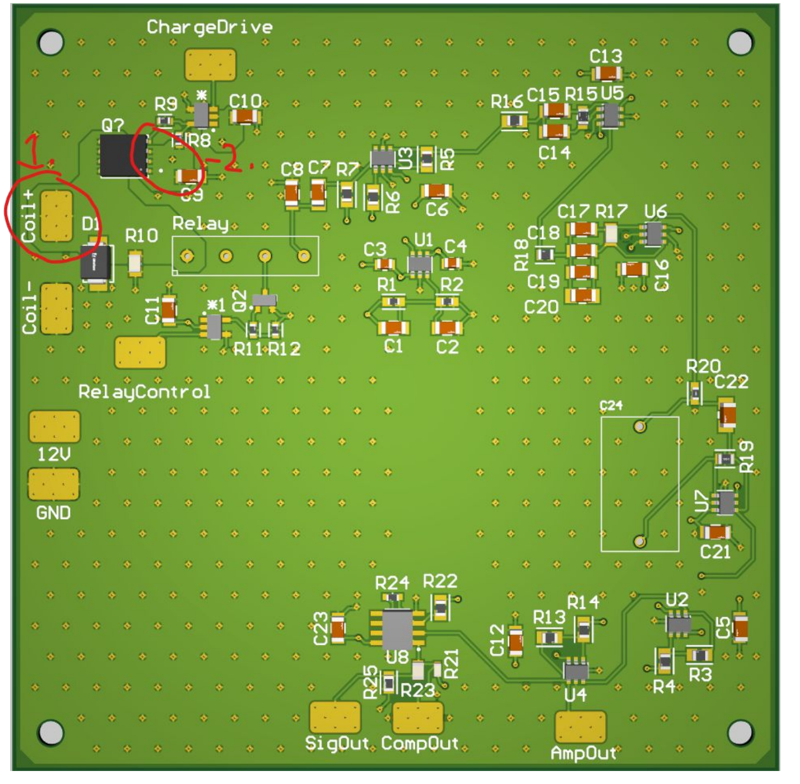
PCB Design Overview

- 4 Layer Board
  1. Signal Layer
  2. GND Layer
  3. 12V Layer
  4. Signal Layer
  - Separated by sections
  - Pads used for wire connections
  - Stitching Vias for GND plane connections
  - [PCB Link]



PCB Issue Info

- Due to error when ordering
  1. Coil+ is not connected to correct pour
  2. N-channel Mosfet source pour not connected to 12V
- These issues were fixed on board by removing solder mask and making connections.
- PCB and schematic are fixed



PCB BOM

PCB Bom

- Link to PCB BOM GitHub: [Link]

PCB1.PcbDoc | BIGPCBV1.BomDoc | Home Page | PowerFiltering.SchDoc | SwitchingCircuit.SchDoc | BufferAmp.SchDoc | BandpassV2.SchDoc | InitialAmp.SchDoc

Group by | Add new | Reset Supply Chain | Failed to pull data | Refresh | Search

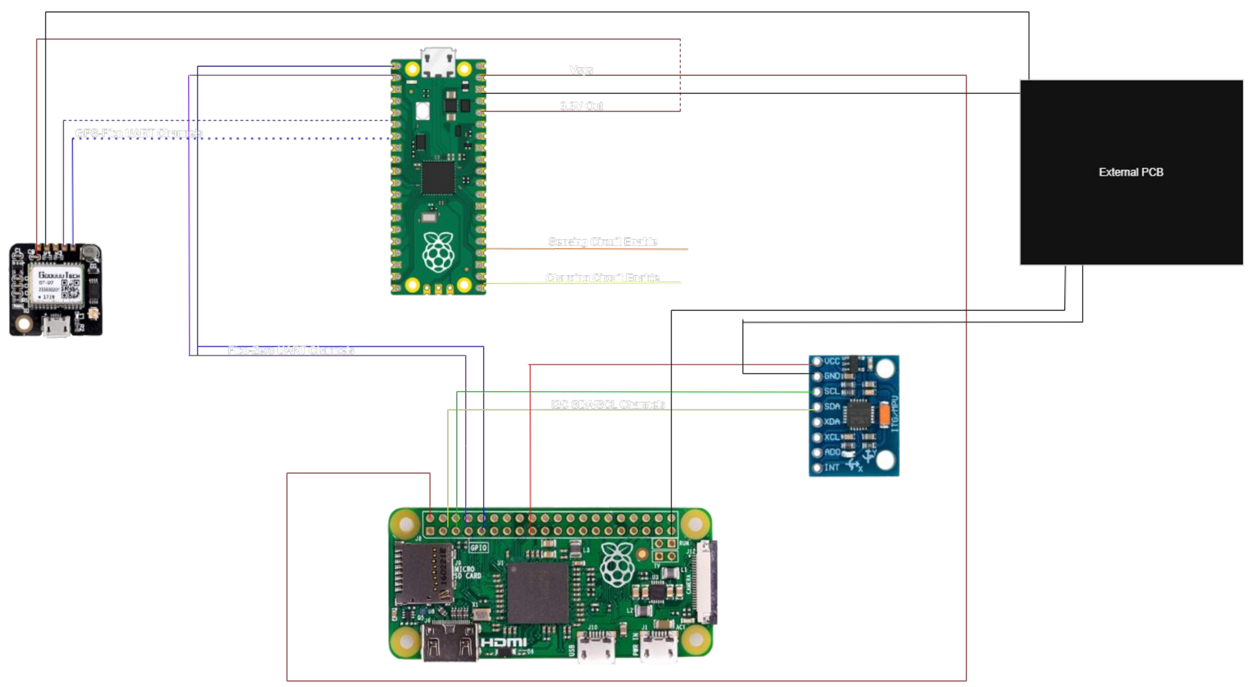
	Item Details									
	Line #	Name	Description	Designatc	Revision ID	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufactur
1		MCP1416T-E/OT	IC GATE DRVR LOW-SIDE SOT23-5 Low-Side Gate Driver IC...	[NoValue], *1			2	Microchip	MCP1416T-E/OT	Volume Prc
2		3570-1331-121	Relay Reed Sip SPST .5A 12V	3570-1331...			1	Comus	3570-1331-121	
3		100n, 100 nF	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C1, C2, C5,...			13	Murata	GRM3195C1H104JA05D	Volume Prc
4		10u	Chip Multilayer Ceramic Capacitors for General Purpose, 080...	C3, C4			2	Murata	GRM21BR71A106KA73L	Volume Prc
5		330nF	CAP CER 0.33UF 25V C0G/NP0 1206	C7			1	Murata	GRM31C5C1E334JE01L	Volume Prc
6		GRM31M5C1H563JA01...	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C8, C15			2	Murata	GRM31M5C1H563JA01L	Volume Prc
7		27 nF	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C14			1	Murata	GRM3195C1H273JA01D	Volume Prc
8		68 nF	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C18, C19			2	Murata	GRM3195C1H683JA05D	Volume Prc
9		47 nF	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C20			1	Murata	GRM3195C1H473JA05D	Volume Prc
10		75 nF	Chip Multilayer Ceramic Capacitors for General Purpose, 120...	C22			1	Murata	GRM31C5C1H753JA01L	Volume Prc
11		ECWFD2W185K		C24			1			
12		SMBJ6.0CA	TVS DIODE 6V 10.3V DO214AA	D1			1	Littelfuse	SMBJ6.0CA	Volume Prc
13		NX7002AK,215	Single N-Channel Trench MOSFET, 60 V, -55 to 150 degC, 3-...	Q2			1			
14		SIR873DP-T1-GE3	P-CHANNEL 150-V (D-S) MOSFET	Q?			1			
15		100k	Automotive Metal Thin Film Chip Resistor, 0603, 100kΩ, 0.1...	R1, R2			2	Susumu	RG1608P-104-B-T5	Volume Prc
16		10k	10kΩ ±0.5% 0.1W 1/10W Chip Resistor 0805 (2012 Metric)...	R3, R5, R7,...			4	Susumu	RR1220P-103-D	Not Recom
17		1k	1kΩ ±0.5% 0.1W 1/10W Chip Resistor 0805 (2012 Metric) T...	R4, R6, R14...			4	Susumu	RR1220P-102-D	Not Recom
18		10k	10kΩ ±1% 0.1W 0603 Thick Film Chip Resistor AEC-Q200 c...	R8, R12			2	Stackpole Electr...	RMCF0603FT10K0	Volume Prc
19		10	10Ω ±1% 0.1W 0603 Thick Film Chip Resistor AEC-Q200 co...	R9, R11			2	Stackpole Electr...	RMCF0603FT10R0	Volume Prc
20		24.9k		R10			1	Vishay	CRCW080524K9FKEA	Volume Prc
21		1.87 k	RES SMD 1.87K OHM 0.1% 1/8W 0805	R15			1	Yageo Group	RT0805BRD071K87L	Volume Prc
22		270	270Ω ±0.5% 0.1W 1/10W Chip Resistor 0805 (2012 Metric)...	R16			1	Susumu	RR1220P-271-D	Not Recom
23		3.90 k		R17			1	Panasonic	ERA6AEB392V	
24		562	562Ω ±1% 0.25W 0805 High Power Anti-Sulfur Thin Film Ch...	R18			1	Stackpole Electr...	RNCP0805FTD562R	Volume Prc
25		442	442Ω ±0.1% 0.125W, 1/8W Chip Resistor 0805 (2012 Metri...	R19			1	KOA Speer	RN73R2ATTD4420B05	Volume Prc
26		94.2	94.2Ω ±0.1% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric)...	R20			1	KOA Speer	RN73R1JTTD94R2B25	Volume Prc
27		4.7k	Chip Resistor, 4.7 KOhm, +/- 1%, 0.1 W, -55 to 155 degC, 0...	R21			1	Yageo Group	RC0603FR-074K7L	Volume Prc
28		270k	270 kOhms ±0.1% 0.125W, 1/8W Chip Resistor 0805 (2012...	R23			1	Panasonic	ERA-6AEB274V	Volume Prc
29		1.21M	RES 1.21M OHM 1% 1/10W 0603	R24			1	Vishay	CRCW06031M21FKEA	Volume Prc
30		100k	RES 100K OHM 0.1% 1/8W 0805	R25			1	Panasonic	ERA-6AEB104V	
31		LT6236CS6TRMPBF	Rail-to-Rail Output 215MHz, 1.1nV/âˆšHz Op Amp/ SAR AD...	U1, U2, U3...			7			
32		LM393DR2G	Low Offset Voltage Dual Comparators, 2 to 36 V, 0 to 70 de...	U8			1	onsemi	LM393DR2G	Volume Prc

32 of 32 lines visible

Software

High-Level Overview

- Two Microcontrollers at use – Pi Zero and Pi Pico which communicate via UART
- Peripherals include GT-U7 GPS Module and a GY-521 IMU



Why 2 Microcontrollers?

- Please trust me, if you want to expand features/functionality to the boat keep 2 microcontrollers
- Pi Pico has 2 UART channels, but if you axe the pi zero, you can only read UART from GPS and one more peripheral, but still need to communicate with the boat, write to an external SD etc.
- Pi Zero also cannot be sole microcontroller (for the time being), as it doesn’t have the PIO functionality onboard the RP2040 Pico chip used for measuring the comparator output.

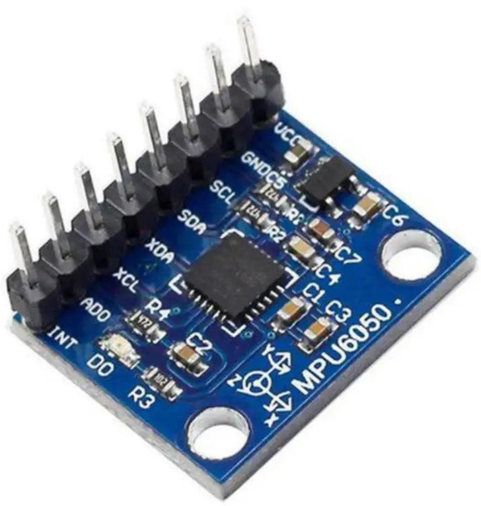
Division of Labor Between Microcontrollers

- **Pi Pico Responsibilities :**
- Use PIO to gather frequency data fed from coil comparator (initiated by Pi Zero)
- Read GT-U7 GPS coordinates
- Send data VIA UART to PI

- **Pi Zero Responsibilities :**
- Send controls signals to both send current through coils and stop charging
- Tell Pi Zero via UART to begin measurement
- Receive GPS/Frequency Measurements, apply IMU corrections to frequency data and calculate magnetic field strength
- Format all data into CSV file for storage, to be uncovered after a completed journey.

GY-521 IMU

- Need to account for roll, pitch and yaw of the Towfish when measuring magnetic field in a single dimension
- Can gather this data by polling an IMU for duration of frequency measurement, and applying corrections
- Calculations and extra things of that nature are in a LaTeX file inside of the github repo – it's like 3 pages of calculations that are not needed here.



Using Programable IO (PIO) to Measure Sensor Comparator Frequency

- Programmable IO (PIO) is a feature new to RP2040 chip, allowing for low-level assembly-like control of IO ports
- Take advantage of this instead of unreliable pi pico ADC in order to poll one pin from a comparator at a high frequency
- Can read 9MHz reliably from a lab oscilloscope, reads a 2kHz signal (frequency of Earths's b field) at a 0.3nT sensitivity
- Code is in the Github repo – please use it, it works well!
- If you'd like to axe the comparator and do an ADC, I'd recommend using a CPLD, an altera one preferably so you can use modelsim ,quartus etc tools that UW students are familiar with.

Turning Frequency into Magnetic Field and Sensitivity

The system converts measured frequency into magnetic field strength and estimates stability.

Frequency to magnetic field:

$$\omega = \gamma_p B, \quad \omega = 2\pi f$$

$$B = \frac{2\pi f}{\gamma_p} \quad \Rightarrow \quad B_{\mathrm{nT}} = \left(\frac{2\pi f}{\gamma_p}\right) \cdot 10^9$$

Discrete magnetic field series:

$$B_i = \frac{2\pi f_i}{\gamma_p}$$

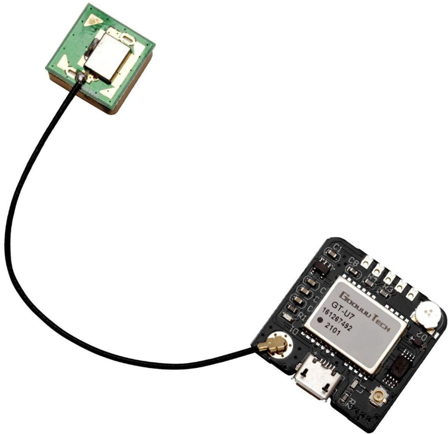
Fourth-order sensitivity estimator:

$$T_i = B_{i-2} - 4B_{i-1} + 6B_i - 4B_{i+1} + B_{i+2}$$

$$\bar{T} = \frac{1}{M} \sum T_i$$

GT-U7 GPS Module

- Gather GPS coordinates and send via UART channels to Pi Pico.
- GT-U7 takes a while to lock, module must have access to the sky to lock onto satellites.
- For future groups that will integrate this into telemetry of the boat – there are boat versions of the code with software that will enable idle charging and will tell the boat not to move until a gps-lock is achieved. Both of these require transmitting GPS coordinate to the boat's microcontroller via UART or some other comm. Protocol.



IMPORTANT

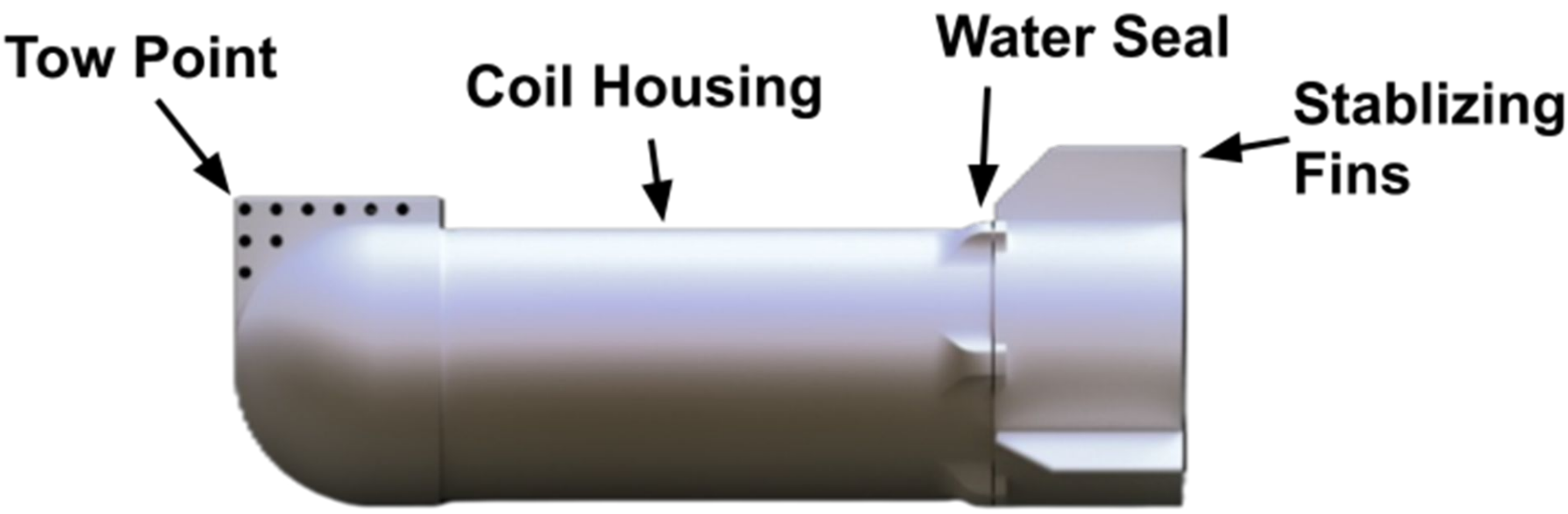
- The closer the timing window between the charging circuit discharging and initiating the sensing circuit, the better the measurement
- However – if the charge circuit is still discharging and the sensor circuit is enabled there is a chance of damage to the sensing circuit.

Physical Towfish Design

Physical TowFish Design

- Low-cost 3D printed housing allows for more customization
- Multiple cable attachment points
- Hydro stabilizing fins

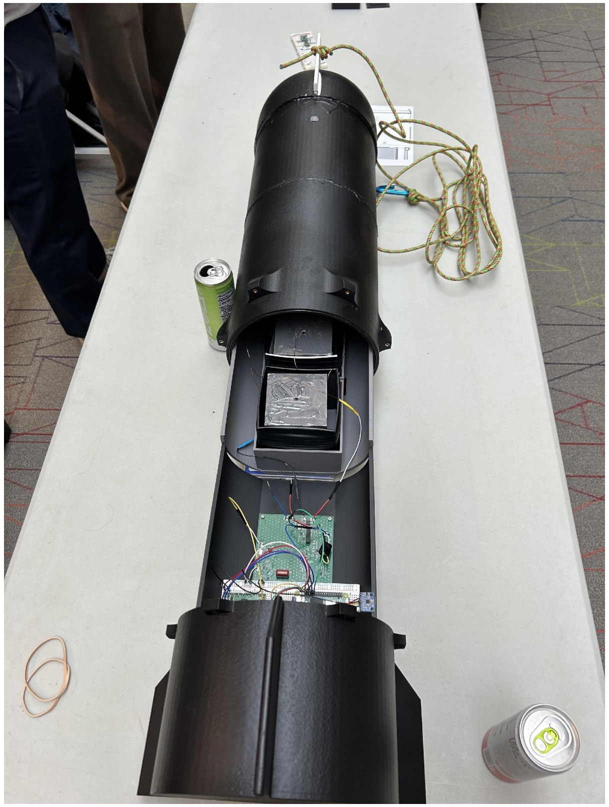
- 8 inch diameter housing to store large coils + proton rich fluid
- End cap complete with O-ring to provide waterproof seal



Physical Towfish Design Process

- Glued in stopper block at front edge of Front Tube piece with Super Glue
- Glued in O-ring to Back Tube with super glue
  - Super Glue
  - O'ring
- Connected Nose Cone to Front Tube and Front Tube to Back Tube with JB Weld Marine Weld
  - J-B WELD Epoxy Adhesive: MarineWeld
- Coated Entire Outside of all sections with Flex Seal Rubber Sealant to make 3D Prints Waterproof
  - Flex Seal Liquid Rubber Sealant Coating Spray

Physical TowFish Design Process



Physical TowFish Design Documentation Links

- 3D Model of Nose Cone GitHub Link:
  - Link
- 3D model of Front Tube GitHub Link:
  - Link
- 3D Model of Back Tube GitHub Link:
  - Link
- 3D Model of Back Cap GitHub Link:
  - Link
- 3D Model of Coil Holder inside of TowFish:
  - Link
- 3D Model of PCB Holder inside of TowFish:
  - Link
- 3D Model of Stopper Block inside of TowFish:
  - Link

General Notes

General Notes

- Simulation files for most circuit schematics are available on GitHub to demonstrate working designs
- Stencils for spray-painting Towfish with a Wisconsin inspired design have been added to GitHub
- Finalize Towfish waterproofing by inserting the cable into the cable hole in Front Tube and seal with with JB Weld

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